REPUBLIC OF THE PHILIPPINES
BICOL UNIVERSITY
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BAGONG PILIPINAS

IT112 Web System Development

Phase 2 – Backend Development

Lab Report by TechNet

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I. Overview of Backend Setup

The **TaskWise Backend** follows a modular, feature-based architecture using Node.js, Express, and MongoDB. It is designed for scalability, maintainability, and clear separation of concerns.

II. Code Snippets

A. server.js

```
const express = require("express");
require("dotenv").config();
const cors = require("cors");
const helmet = require("helmet");

const connectDB = require("./config/db");

const studentRouter = require("./modules/routes/student.router");
const groupRouter = require("./modules/routes/group.router");
const taskRouter = require("./modules/routes/task.router");

const app = express();

const routers = [studentRouter, groupRouter, taskRouter];

app.use(express.json());
app.use(helmet());
app.use(cors());

routers.forEach((router) => {
  app.use("/api", router);
});

async function start() {
  const port = process.env.PORT;
  const uri = process.env.MONGO_URI_LOCAL;

  await connectDB(uri);

  app.listen(port, () => {
    console.log(`Server running at http://localhost:${port}`);
  });
}

start();
```

B. Sample Routes – task.router.js

```
const express = require("express");

const taskRouter = express.Router();

const {
  getTasks,
  createTask,
  updateTask,
  deleteTask,
} = require("../controllers/task.controller");

const { setPriority } =
require("../controllers/task.priority.controller");
const assignTask =
require("../controllers/task.assignment.controller");
const authMiddleware = require("../middlewares/auth.middleware");

// GET ALL THE TASKS
taskRouter.get("/task", authMiddleware, getTasks);

// CREATE TASK
taskRouter.post("/task/create", authMiddleware, createTask);

// UPDATE TASK
taskRouter.patch("/task/update/:id", authMiddleware, updateTask);

// TASK ASSIGNMENT
taskRouter.post("/task/assign", authMiddleware, assignTask);

// TASK PRIORITY
taskRouter.post("/task/priority/:id", authMiddleware,
setPriority);

// DELETE TASK
taskRouter.delete("/task/delete/:id", authMiddleware,
deleteTask);

module.exports = taskRouter;
```

C. Sample Controller – task.controller.js

```
const taskService = require("../services/task.service");

const getTasks = async (req, res) => {
```

```
  const { id } = req.student;
  try {
    const tasks = await taskService.getTasksByCreator(id);

    res.status(200).json({ tasks });
  } catch (error) {
    res.status(500).json({ message: error.message });
  }
};

const createTask = async (req, res) => {
  const task = req.body;
  const { id } = req.student;
  try {
    const newTask = await taskService.createTask(task, id);
    res.status(201).json({ newTask });
  } catch (error) {
    res.status(500).json({ message: error.message });
  }
};

const updateTask = async (req, res) => {
  const { id } = req.student;
  try {
    const newTask = await taskService.updateTask(req.params.id,
id, req.body);
    res.status(200).json({ newTask });
  } catch (error) {
    res.status(500).json({ message: error.message });
  }
};

const deleteTask = async (req, res) => {
  const { id } = req.student;
  try {
    const task = taskService.deleteTask(req.params.id, id);

    res.status(200).json({ task });
  } catch (error) {
    res.status(400).json({ message: error.message });
  }
};

module.exports = { getTasks, createTask, updateTask, deleteTask
};
```

D. Sample Service – task.service.js

```
const {
  create,
  findByCreator,
  findByIdAndCreator,
  updateById,
  deleteById,
} = require("../repositories/task.repo");

const getTasksByCreator = async (id) => {
  return await findByCreator(id);
};

const createTask = async (task, id) => {
  return await create(task, id);
};

const updateTask = async (taskId, studentId, taskInfo) => {
  const allowedFields = ["name", "difficulty", "deadline"];
  const task = {};

  for (let key of allowedFields) {
    if (taskInfo[key] !== undefined) {
      task[key] = taskInfo[key];
    }
  }

  const foundedTask = await findByIdAndCreator(taskId,
studentId);

  if (!foundedTask) throw new Error("Unauthorized");

  const newTask = await updateById(taskId, task);

  if (!newTask) {
    return res.status(404).json({ message: "Task not found" });
  }

  return newTask;
};

const deleteTask = async (taskId, studentId) => {
  const task = await findByIdAndCreator(taskId, studentId);
  if (!task) throw new Error("Unauthorized");
```

```
  await deleteById(taskId);
  return task;
};

module.exports = { getTasksByCreator, createTask, updateTask,
deleteTask };
```

E. Sample Repository – task.repo.js

```
const Task = require("../models/task.model");

const findByCreator = async (creator) => {
  return await Task.find({ creator });
};

const create = async (taskInfo, creator) => {
  return await Task.create({
    ...taskInfo,
    creator,
  });
};

const findByIdAndCreator = async (_id, creator) => {
  return await Task.findOne({ _id, creator });
};

const updateById = async (taskId, task) => {
  return await Task.findByIdAndUpdate(
    taskId,
    { $set: task },
    { new: true, runValidators: true },
  );
};

const deleteById = async (taskId) => {
  return await Task.findByIdAndDelete(taskId);
};

module.exports = {
  findByCreator,
  create,
  findByIdAndCreator,
  updateById,
  deleteById,
};
```

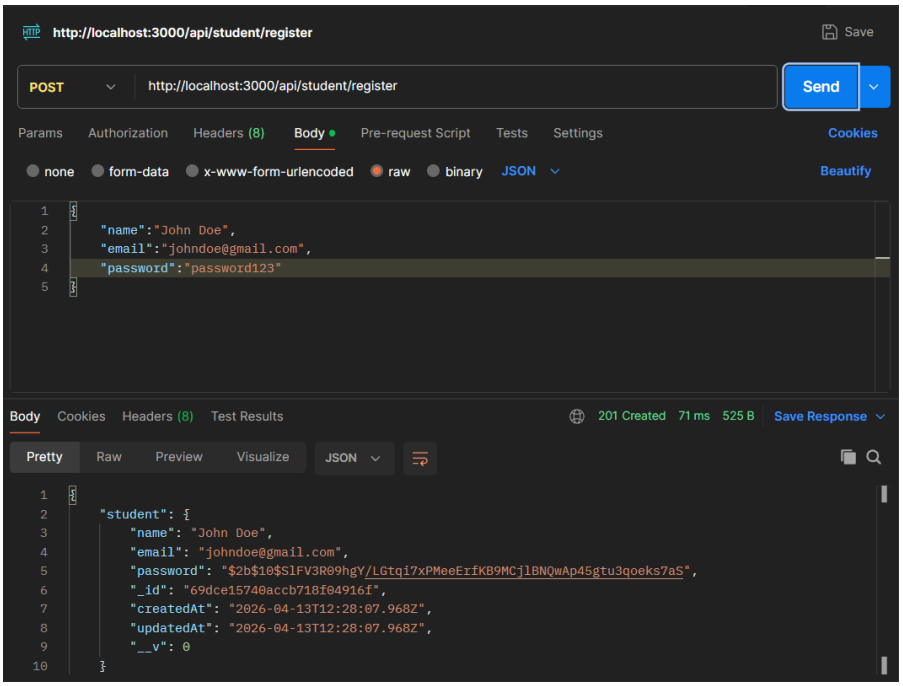
F. Sample Model – task.model.js

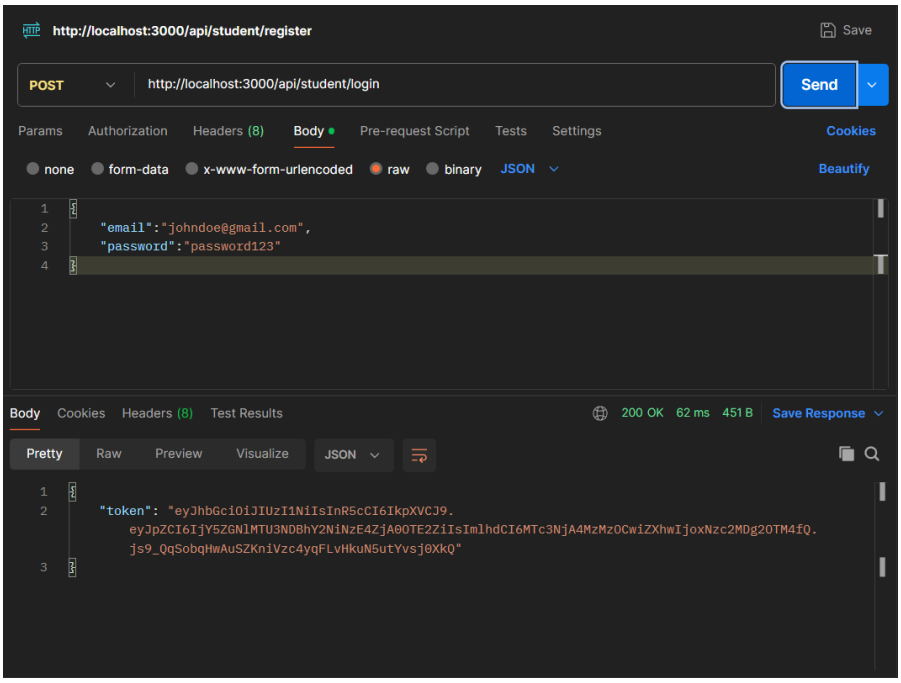
```
const mongoose = require("mongoose");
const taskSchema = new mongoose.Schema(
  {
    name: {
      type: String,
      maxlength: 20,
    },
    difficulty: {
      type: Number,
      enum: [1, 2, 3, 4, 5],
    },
    deadline: {
      type: Date,
      required: true,
    },
    subject: {
      type: String,
      maxlength: 20,
    },
    creator: {
      type: mongoose.Schema.Types.ObjectId,
      ref: "Student",
    },
  },
  { timestamps: true },
);

const Task = mongoose.model("Task", taskSchema);

module.exports = Task;
```

III. API Testing

All routes were thoroughly tested using Postman, and the corresponding test cases are documented below.

A. Student





B. Task

http://localhost:3000/api/tasks

Save

GET http://localhost:3000/api/task Send

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary

This request does not have a body

Body Cookies Headers (8) Test Results 200 OK 12 ms 799 B Save Response

Pretty Raw Preview Visualize JSON


```
{
    "tasks": [
        {
            "_id": "69d64e429c22d63b8266c21f",
            "name": "Programming",
            "difficulty": 2,
            "deadline": "2026-04-22T16:00:00.000Z",
            "subject": "Computer Programming",
            "creator": "69d36694f23ea91eccf967b9",
            "createdAt": "2026-04-08T12:46:58.910Z",
```


http://localhost:3000/api/task/create

Save

POST http://localhost:3000/api/task/create Send

Params Authorization Headers (9) Body Pre-request Script Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary JSON Beautify


```
{
    "name":"Test",
    "difficulty":1,
    "deadline":"04-27-26",
    "subject":"Mathematics"
}
```


Body Cookies Headers (8) Test Results 201 Created 16 ms 534 B Save Response

Pretty Raw Preview Visualize JSON


```
{
    "newTask": {
        "name": "Test",
        "difficulty": 1,
        "deadline": "2026-04-26T16:00:00.000Z",
        "subject": "Mathematics",
        "creator": "69d36694f23ea91eccf967b9",
        "_id": "69dce29140accb718f049176",
        "createdAt": "2026-04-13T12:33:21.877Z",
        "updatedAt": "2026-04-13T12:33:21.877Z",
```

http://localhost:3000/api/task/create

Save

PATCH http://localhost:3000/api/task/update/69dce29140accb718f049176 Send

Params Authorization Headers (9) Body Pre-request Script Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary JSON Beautify

```
{
    "difficulty":4
}
```

Body Cookies Headers (8) Test Results 200 OK 22 ms 529 B Save Response

Pretty Raw Preview Visualize JSON


```
{
    "newTask": {
        "_id": "69dce29140accb718f049176",
        "name": "Test",
        "difficulty": 4,
        "deadline": "2026-04-26T16:00:00.000Z",
        "subject": "Mathematics",
        "creator": "69d36694f23ea91eccf967b9",
        "createdAt": "2026-04-13T12:33:21.877Z",
        "updatedAt": "2026-04-13T12:35:02.928Z",
```


http://localhost:3000/api/task/assign

Save

POST http://localhost:3000/api/task/assign Send

Params Authorization Headers (9) Body Pre-request Script Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary JSON Beautify

```
{
    "taskId":"69df556dd61633f02d6f5dcd",
    "assignedTo":"69d3616950f737dbfd0b61a0",
    "groupId":"69df67b30996d34f6eb3fc8e"
}
```

Body Cookies Headers (19) Test Results 201 Created 74 ms 1.17 KB Save Response

Pretty Raw Preview Visualize JSON


```
{
    "assigned": {
        "taskId": "69df556dd61633f02d6f5dcd",
        "assignedTo": "69d3616950f737dbfd0b61a0",
        "groupId": "69df67b30996d34f6eb3fc8e",
        "_id": "69e0bdab15c7360fb33a9307",
        "createdAt": "2026-04-16T10:44:59.014Z",
        "updatedAt": "2026-04-16T10:44:59.014Z",
        "__v": 0
    },
```

http://localhost:3000/api/task/priority/69dce26e40accb718f049174

Save

POST http://localhost:3000/api/task/priority/69df556dd61633f02d6f5dcd Send

Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary JSON Beautify

Body Cookies Headers (19) Test Results 201 Created 21 ms 1.08 KB Save Response

Pretty Raw Preview Visualize JSON


```
{
    "priorityLevel": {
        "taskId": "69df556dd61633f02d6f5dcd",
        "priorityLevel": "LOW",
        "_id": "69e0bee915c7360fb33a930c",
        "createdAt": "2026-04-16T10:50:17.343Z",
        "updatedAt": "2026-04-16T10:50:17.343Z",
        "__v": 0
    }
}
```


http://localhost:3000/api/task/delete/69dce29140accb718f049176

Save

DELETE http://localhost:3000/api/task/delete/69dce6b40d21f1932a7824bd Send

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

Query Params

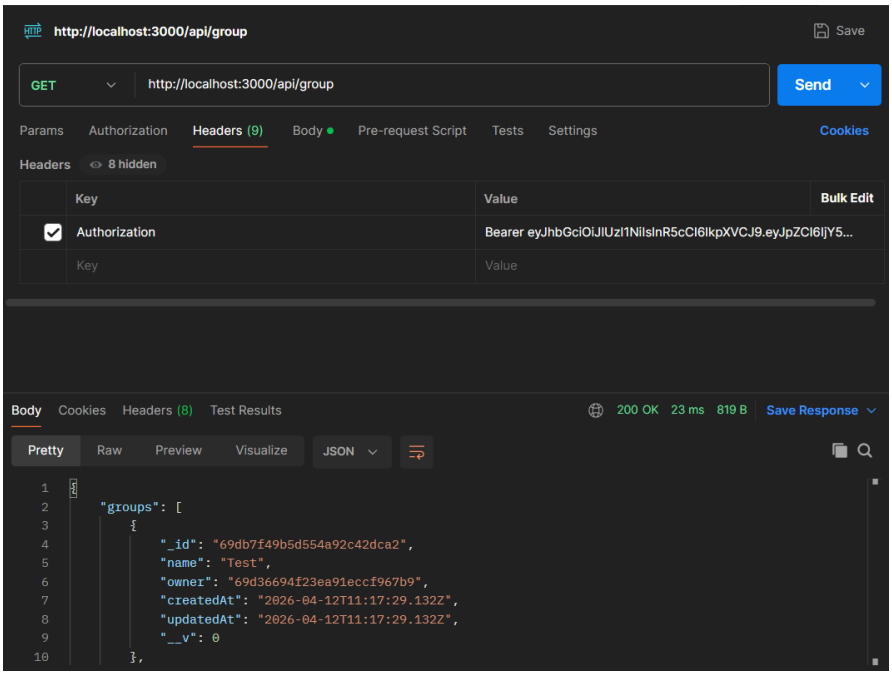
Key	Value	Bulk Edit
Key	Value	

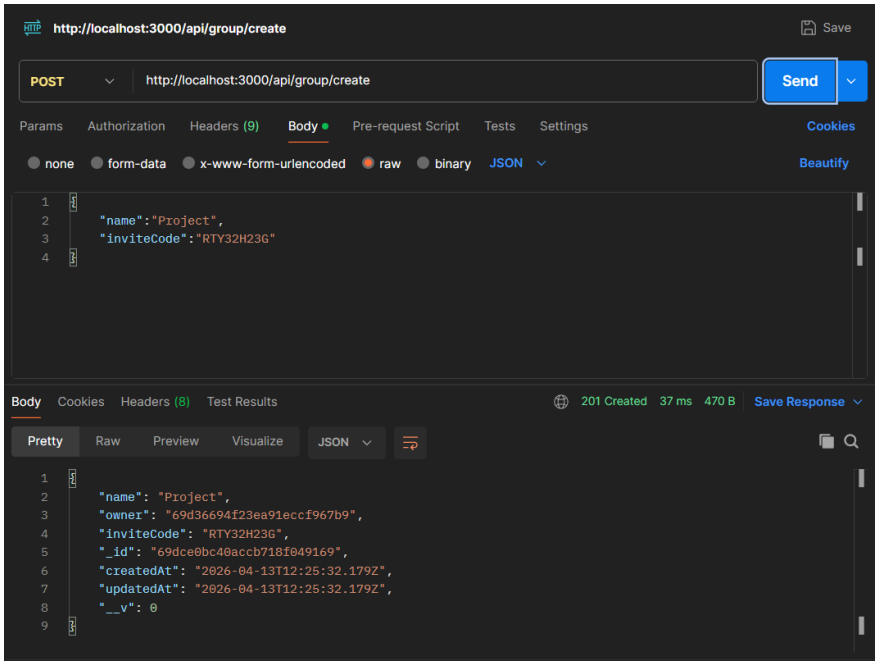
Body Cookies Headers (8) Test Results 200 OK 29 ms 533 B Save Response

Pretty Raw Preview Visualize JSON


```
{
    "deletedTask": {
        "_id": "69dce6b40d21f1932a7824bd",
        "name": "Test",
        "difficulty": 1,
        "deadline": "2026-04-26T16:00:00.000Z",
        "subject": "Mathematics",
        "creator": "69d36694f23ea91eccf967b9",
        "createdAt": "2026-04-13T12:51:00.311Z",
        "updatedAt": "2026-04-13T12:51:00.311Z",
```

C. Group


http://localhost:3000/api/group
Save
GET
http://localhost:3000/api/group
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
Headers
8 hidden
Key
Value
Bulk Edit
Authorization
Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5...
Key
Value
Body
Cookies
Headers (8)
Test Results
200 OK
23 ms
819 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
{
"groups": [
{
"_id": "69db7f49b5d554a92c42dca2",
"name": "Test",
"owner": "69d36694f23ea91eccf967b9",
"createdAt": "2026-04-12T11:17:29.132Z",
"updatedAt": "2026-04-12T11:17:29.132Z",
"__v": 0
},


http://localhost:3000/api/group/create
Save
POST
http://localhost:3000/api/group/create
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
{
"name":"Project",
"inviteCode":"RTY32H23G"
}
Body
Cookies
Headers (8)
Test Results
201 Created
37 ms
470 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
{
"name": "Project",
"owner": "69d36694f23ea91eccf967b9",
"inviteCode": "RTY32H23G",
"_id": "69dce0bc40accb718f049169",
"createdAt": "2026-04-13T12:25:32.179Z",
"updatedAt": "2026-04-13T12:25:32.179Z",
"__v": 0
}

http://localhost:3000/api/group/create

Save

POST http://localhost:3000/api/group/join Send

Params Authorization Headers (9) Body Pre-request Script Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary JSON Beautify

```
{
  "inviteCode":"RTY32H23G"
}
```

Body Cookies Headers (8) Test Results 200 OK 28 ms 491 B Save Response

Pretty Raw Preview Visualize JSON

```
{
    "member": {
        "groupId": "69db7eba19066b9befa2ba5e",
        "studentId": "69d36694f23ea91eccf967b9",
        "role": "member",
        "_id": "69dce0ec40accb718f04916d",
        "createdAt": "2026-04-13T12:26:20.320Z",
        "updatedAt": "2026-04-13T12:26:20.320Z",
        "__v": 0
    }
```

http://localhost:3000/api/group/delete/69db7eba19066b9befa2ba5e

Save

DELETE http://localhost:3000/api/group/delete/69db7f7fb5d554a92c42dca5 Send

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

Query Params

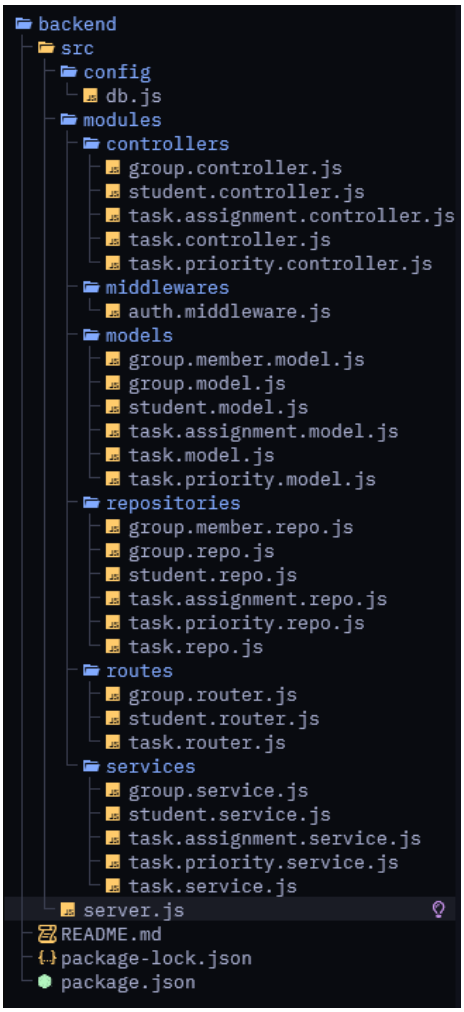
Key	Value	Bulk Edit
Key	Value	

Body Cookies Headers (8) Test Results 200 OK 28 ms 457 B Save Response

Pretty Raw Preview Visualize JSON

```
{
    "deletedGroup": {
        "_id": "69db7f7fb5d554a92c42dca5",
        "name": "Project",
        "owner": "69d36694f23ea91eccf967b9",
        "createdAt": "2026-04-12T11:18:23.516Z",
        "updatedAt": "2026-04-12T11:18:23.516Z",
        "__v": 0
    }
}
```

IV. Backend Folder Structure



V. Group Member Contributions

Member	Role	Contribution
Jay Lloyd Chicogo	Backend Developer	He built and maintains the server side of the system by setting up the Express server, creating routes, controllers, and models, and ensuring that all data processing and API responses work correctly and

		align with the needs of the frontend.
John Nelson Andes	Database Manager	He manages the database by creating a Mongoose models, establishing and maintaining the MongoDB connection, and ensuring that all data is properly structured, stored, and retrieved in a reliable and efficient way.
Jay Lloyd Chicogo	Github Manager	His job is to oversee the project repository by organizing the folder structure, commiting code consistently with clear and meaningful messages, and ensuring that all contributions are properly tracked and synchronized among team members.
Mico Malana	Documentation Officer	He handled the project documentation by capturing screenshots, explaining the functionality of each route, and preparing a clear and organized lab report that accurately reflects the system and its features.
Mico Malana	Project Manager	Monitors the overall progress of theproject by reviewing each assigned task, ensuring that all contributions are aligned with the project goals and requirements, and checking that the team stays on track with deadlines and expected outcomes.